INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING(IAIML)

R22-SECTION-A,B,C

- UNIT-1 INTRODUCING ARTIFICIAL INTELLIGENCE
- UNIT-2:UNDERSTANDING MACHINE LEARNING
- UNIT-3:LOOKING INSIDE MACHINE LEARNING
- UNIT-4:TYPES OF MACHINE LEARNING ALGORITHMS AND DATA CONTINUM
- UNIT-5:TOOLS OF AIML

SYLLABUS

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(22ES1AM101) INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TEACHING SCHEME		
L	T/P	C
2	0	2

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the concepts of Artificial Intelligence and its applications
- To introduce the concepts of Machine learning and its applications
- To understand the various AIML algorithms and illustrate machine learning cycle
- To learn types of machine learning algorithms and data continuum
- To explore various tools for AIML

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explain the definition of Artificial Intelligence and its applications
CO-2: Elucidate the definition of Machine Learning and its applications
CO-3: Explicate various AIML algorithms and machine learning cycle
CO-4: Discuss machine learning algorithms and data continuum
CO-5: Explore various tools for AIML.

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	1	1	-	-	1	-	-	-	1	-	-	3	3	2
CO-2	2	-	2	1	3	-	1	-	-	-	-	-	3	2	2
CO-3	2	1	-	1	2	1	-	-	-	1	-	-	2	3	2
CO-4	2	1	2	-	2	-	1	-	-	1	-	-	2	2	2
CO-5	2	2	2	1	2	1	1	-	-	1	-	-	-	-	-

UNIT-I:

Introducing Artificial Intelligence: Introducing AI, Definition of AI, understanding kinds of intelligence, discovering 4 ways to define AI, History of AI.

UNIT-II:

Understanding ML: Machine Learning definition, Approaches to ML: Supervised, Unsupervised, Reinforcement learning, Neural Networks & Deep Learning.

UNIT-III:

Looking Inside ML: The Impact of Machine Learning on Applications: Role of algorithms, The Machine Learning Cycle, Issues in Machine Learning.

UNIT-IV:

Types of Machine Learning Algorithms and Data Continuum: Types of machine learning algorithms, Training machine learning systems, identifying relevant data: Structured data sources and Unstructured data sources.

UNIT-V:

Tools in ML: Introduction to AI Tools: Python, R, MatLab, SAS, Other Tools.

TEXT BOOKS:

1. Artificial Intelligence For Dummies (For Dummies (Computer/Tech)), John Paul Mueller, Luca Massaron, Wiley, 2018
2. Machine Learning for Dummies, Judith Hurwitz and Daniel Kirsch, IBM Limited Edition, A. Wiley, 2018
3. Machine Learning, Saikat Dutt, Pearson, 1st Edition, 2018

REFERENCES:

1. Machine Learning (in Python and R) For Dummies, John Paul Mueller
2. Artificial Intelligence Basics: A Non-Technical Introduction, Tom Taulli, 2019
3. Machine Learning for Absolute Beginners: A Plain English Introduction, O. Theobald, 2nd Edition

ONLINE RESOURCES:

1. https://www.softwaretestinghelp.com/machine-learning-tools/
2. https://intellipaat.com/blog/top-artificial-intelligence-tools/

EXAM PATTERN

- Each semester shall have ‘Continuous Internal Evaluation (CIE)’ and ‘Semester End Examination (SEE).
- A student shall maintain a minimum required attendance of 75% in AGGREGATE to be eligible to appear for the Semester End Examinations.
- A minimum of 35% marks (14 marks out of 40 marks) for each course in the continuous internal evaluation (CIE)

14.0 MINIMUM ACADEMIC REQUIREMENTS:

- In addition to the attendance requirements mentioned in **13.0** of these regulations for the award of B.Tech. degree, a student must satisfy the minimum academic requirements as given below:
- A student shall be deemed to have satisfied the minimum academic requirements and earned the credits for each **theory, practical, design and drawing** course in B.Tech. programme, if student secures
  - ✓ A **minimum of 35% marks (14 marks out of 40 marks)** for each course in the **continuous internal evaluation (CIE)**
  - ✓ A **minimum of 35% marks (21 marks out of 60 marks)** for each course in the **semester end examination (SEE)**, and
  - ✓ A **minimum of 40% marks (40 marks out of 100 marks)** for each course considering **both CIE and SEE** taken together.
  - ✓ In terms of letter grades, a '**C**' grade and above in that course.
  - ✓ A student is **eligible to write Semester End Examination** of the concerned course, if the **student scores ≥ 35% marks (i.e., 14 marks out of 40 marks) in CIE.**
  - ✓ **In case, the student appears for Semester End Examination (SEE) of the concerned course but fails to score minimum 35% marks in CIE (i.e., 14 marks out of 40 marks), then his/her performance in that course in SEE shall stand cancelled inspite of student appearing for the SEE.**

16.1 Theory Courses:

- The syllabus for the theory courses shall be divided into **FIVE** units and each unit carries equal weightage in terms of marks distribution.
- For theory courses, the distribution of marks shall be **40 marks for Continuous Internal Evaluation (CIE)** and **60 marks for the Semester End Examination (SEE)**.

❖ **Continuous Internal Evaluation (CIE - 40 M):**

- Continuous Internal Evaluation (**CIE**) shall consist of sessional examination (**SE - 30 M**), Class Assessment (**CA - 05 M**) and Experiential Learning Assessment (**ELA – 05 M**).

➢ **Sessional Examination (SE - 30 M):**

- For theory courses, **two sessional examinations (SE-I and SE-II)** shall be conducted in each semester as per the academic calendar. Each sessional examination shall be evaluated for **30 marks.**

- The **time** duration of each SE shall be **2 hours**.
- Question paper pattern for **SE** (**30 Marks**) shall be as follows:

PART-A: 5 X 2 M = 10 M

a. All questions are **compulsory**.
b. 02 questions from Unit-I and Unit-II each and 01 question from first half of Unit-III in First sessional examination (SE-I)
c. 01 question from second half of Unit-III and 02 questions from Unit-IV and Unit-V each in Second sessional examination (SE-II)

PART-B: 20 M

a. There shall be a total of 06 questions.
b. There shall be **two questions** from **each UNIT** with **internal choice** i.e., **'either' 'or'** choice.
c. Student shall answer **one question** from **each UNIT**.
d. Part-B Paper pattern is as follows:

Sessional Examination – I (SE-I)	**UNIT-I**	**Q.1** **OR** **Q.2**	**8 M** **8 M**
	UNIT-II	**Q.3** **OR** **Q.4**	**8 M** **8 M**
	UNIT-III	**Q.5** **OR** **Q.6**	**4 M** **4 M**
Sessional Examination – II SE-II)	**UNIT-III**	**Q.1** **OR** **Q.2**	**4 M** **4 M**
	UNIT-IV	**Q.3** **OR** **Q.4**	**8 M** **8 M**
	UNIT-V	**Q.5** **OR** **Q.6**	**8 M** **8 M**

VNRVJIET B.Tech. CBCS R22 Academic Regulations

➢ **Class Assessment (CA – 05 M):**

- **Two class assessments** consisting of any one of the alternative assessment tools (AAT) like **online quiz / assignment / objective exam etc.** shall be conducted covering the syllabus that is completed at the time of conducting the CA and evaluated for **05 marks** each.
- The first and second CAs shall be conducted during the instruction days as per the academic calendar.
- **Average of two CAs** shall be calculated and used as the final class assessment marks.
- The valuation and verification of answer scripts of CIE shall be completed within a week after the conduct of the examination.

➢ **Experiential Learning Assessment (ELA – 05 M):**

- **Course project / course viva-voce / poster presentation / case study** in the concerned course shall be conducted before **Sessional Examination-I (SE-I) and Sessional Examination-II (SE-II)** and evaluated for **05 marks**.
- **Average of two ELAs** shall be calculated and used as the final sessional marks for each course.

❖ **Semester End Examination (SEE – 60 M):**

- The **SEE** shall be conducted at the end of semester for a total of **60 marks.**
- The **time** duration of the SEE shall be **3 hours**.
- The **syllabus** for the **theory courses** shall be divided into **FIVE units** and each unit carries equal weightage in terms of marks distribution.
- Question paper pattern for **SEE** (**60 Marks**) shall be as follows:

  PART-A: 5 X 2 M = 10 M

  a. There shall be **one question from each UNIT**.
  b. All questions are **compulsory**.

  PART-B: 5 X 10 M = 50 M

  a. There shall be **two questions** from **each unit** with **internal choice** i.e., **'either' 'or'** choice.
  b. The student shall **answer one question** from **each UNIT**.
  c. There could be a maximum of two subdivisions in a question i.e., (a) and/or (b).

- Introduction to AI
- Definition of AI
- Understanding various kinds of intelligence
- 4 Ways to define AI
- History of AI

Defining the Term Intelligence/Discerning

Intelligence involves certain mental activities composed of the following activities:

» **Learning:** Having the ability to obtain and process new information.

» **Reasoning:** Being able to manipulate information in various ways.

» **Understanding:** Considering the result of information manipulation.

» **Grasping truths:** Determining the validity of the manipulated information.

»**Seeing relationships:** Divining how validated data interacts with other data.

» **Considering meanings:** Applying truths to particular situations in a manner consistent with their relationship.

» **Separating fact from belief:** Determining whether the data is adequately supported by provable sources that can be demonstrated to be consistently valid

As per the authors mentioned in prescribed text book

Understanding Kinds of Intelligence can be described in following ways:

Sno	Type	Simulation Potential	Human tools	Description
1	Visual Spatial	Moderate	Models,graphs,3D drawings, videos, audios	• Physical environment intelligence such as vacuum cleaners etc., Self driving cars, Vacuum cleaners.
2	Bodily Kinetics	Moderate to high	Specialized equipment and real objects	• Robotics surgeries • Body movements, such as those used by a surgeon or a dancer which requires precision and body awareness. • Computer-assisted intervention for both pre-operative planning and intra-operative guidance.

3	Creative	None	New patterns, inventions Specialized equipment and real objects	• Pattern recognition systems. • An AI can simulate existing patterns of thought and even combine them to create what appears to be a unique presentation but is really just a mathematically based version of an existing pattern. • In order to create, an AI would need to possess self awareness, which would require intrapersonal intelligence.
4	Interpersonal	Low to high	Telephone, audio, videoconference calls, emails, chats	• Search engines etc., • The goal of this form of intelligence is to obtain, exchange, give, and manipulate information based on the experiences of others. • Computers can answer basic questions because of keyword input, not because they understand the question. • Cross-referencing terms in a lookup table and then acting on the instructions provided by the table demonstrates logical intelligence, not interpersonal intelligence

5	Intrapersonal	None	Books, creative material, summary content, books, journals	• Looking inward to understand one's own interests and then setting goals based on those interests is currently a human-only kind of intelligence. • An AI processes numeric input using a set of algorithms and provides an output, is not aware of anything that it does, nor does it understand anything that it does
6	Linguistic	Low	Talking ,oral communication	• Transformers, document exchange etc., • Working with words is an essential tool for communication • This form of intelligence includes understanding spoken and written input, managing the input to develop an answer, and providing an understandable answer as output
7	Logical mathematical	High	Analysis, logical interpretations	• AI focused exploring patterns etc., • Calculating a result, performing comparisons, exploring patterns, and considering relationships are all areas in which computers currently excel.

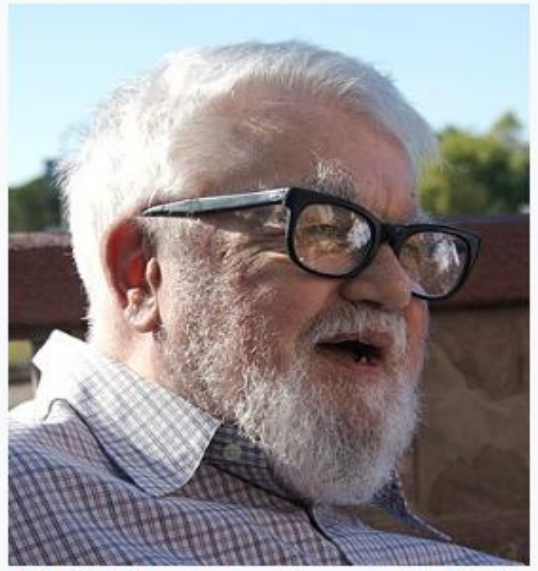
Definition of AI

- AI is understood differently to different people
- AI is defined as the ability of computer systems – Hardware and Software to do tasks, that normally require human intelligence
- AI was coined in the year 1956 by **John Carthy at Darmath College.**
- Eventually Ai is changing the human world.

John McCarthy (September 4, 1927 – October 24, 2011) was an American computer scientist and cognitive scientist. He was one of the founders of the discipline of artificial intelligence.[1] He co-authored the document that coined the term "artificial intelligence" (AI), developed the programming language family Lisp, significantly influenced the design of the language ALGOL, popularized time-sharing, and invented garbage collection.

McCarthy spent most of his career at Stanford University.[2] He received many accolades and honors, such as the 1971 Turing Award for his contributions to the topic of AI,[3] the United States National Medal of Science, and the Kyoto Prize.

John McCarthy



McCarthy at a conference in 2006

Born	September 4, 1927 Boston, Massachusetts, U.S.
Died	October 24, 2011 (aged 84) Stanford, California, U.S.
Alma mater	Princeton University, California Institute of Technology
Known for	Artificial intelligence, Lisp, circumscription, situation calculus
Awards	Turing Award (1971) Computer Pioneer Award (1985 IJCAI Award for Research Excellence (1985)

Definition of AI

- AI is simulation of human intelligence process by machines, especially computer systems.
- AI is the ability for a **computer to think and learn**
- Ai is collection of different technologies working together to **enable machines to sense,** comprehend act and learn with human like levels of intelligence
- AI makes possible for machines to **learn from experience ,**adjust to new inputs and perform human like tasks

History of AI

Year	Milestone
1950	First computing machinery and intelligence was created by Alan Turing
1956	AI term was first coined by John Mc McCarthy at Dartmouth College
1959	First AI Lab was founded at MIT
1974	First AI winter funding Prog got started
1975	MYCIN-was created to diagnose bacterial infections
1987	Second AI winter program began

1989	AI was used in NASA to discover stars
1994	First Web search engine was developed
1997	IBM’s Deep Blue Chess game developed
2002	Amazon replaces human editors with automated system
2011	Apple releases Siri Personal voice agent
2016	Google’s Alpha Go Game was developed

History of AI: For refence purpose

A.I. TIMELINE



1950
TURING TEST
Computer scientist Alan Turing proposes a test for machine intelligence. If a machine can trick humans into thinking it is human, then it has intelligence

1955
A.I. BORN
Term 'artificial intelligence' is coined by computer scientist, John McCarthy to describe "the science and engineering of making intelligent machines"



1961
UNIMATE
First industrial robot, Unimate, goes to work at GM replacing humans on the assembly line



1964
ELIZA
Pioneering chatbot developed by Joseph Weizenbaum at MIT holds conversations with humans

1966
SHAKEY
The 'first electronic person' from Stanford, Shakey is a general-purpose mobile robot that reasons about its own actions

A.I. WINTER
Many false starts and dead-ends leave A.I. out in the cold



1997
DEEP BLUE
Deep Blue, a chess-playing computer from IBM defeats world chess champion Garry Kasparov



1998
KISMET
Cynthia Breazeal at MIT introduces KISmet, an emotionally intelligent robot insofar as it detects and responds to people's feelings



1999
AIBO
Sony launches first consumer robot pet dog AiBO (AI robot) with skills and personality that develop over time



2002
ROOMBA
First mass produced autonomous robotic vacuum cleaner from iRobot learns to navigate and clean homes



2011
SIRI
Apple integrates Siri, an intelligent virtual assistant with a voice interface, into the iPhone 4S



2011
WATSON
IBM's question answering computer Watson wins first place on popular $1M prize television quiz show *Jeopardy*



2014
EUGENE
Eugene Goostman, a chatbot passes the Turing Test with a third of judges believing Eugene is human



2014
ALEXA
Amazon launches Alexa, an intelligent virtual assistant with a voice interface that completes shopping tasks

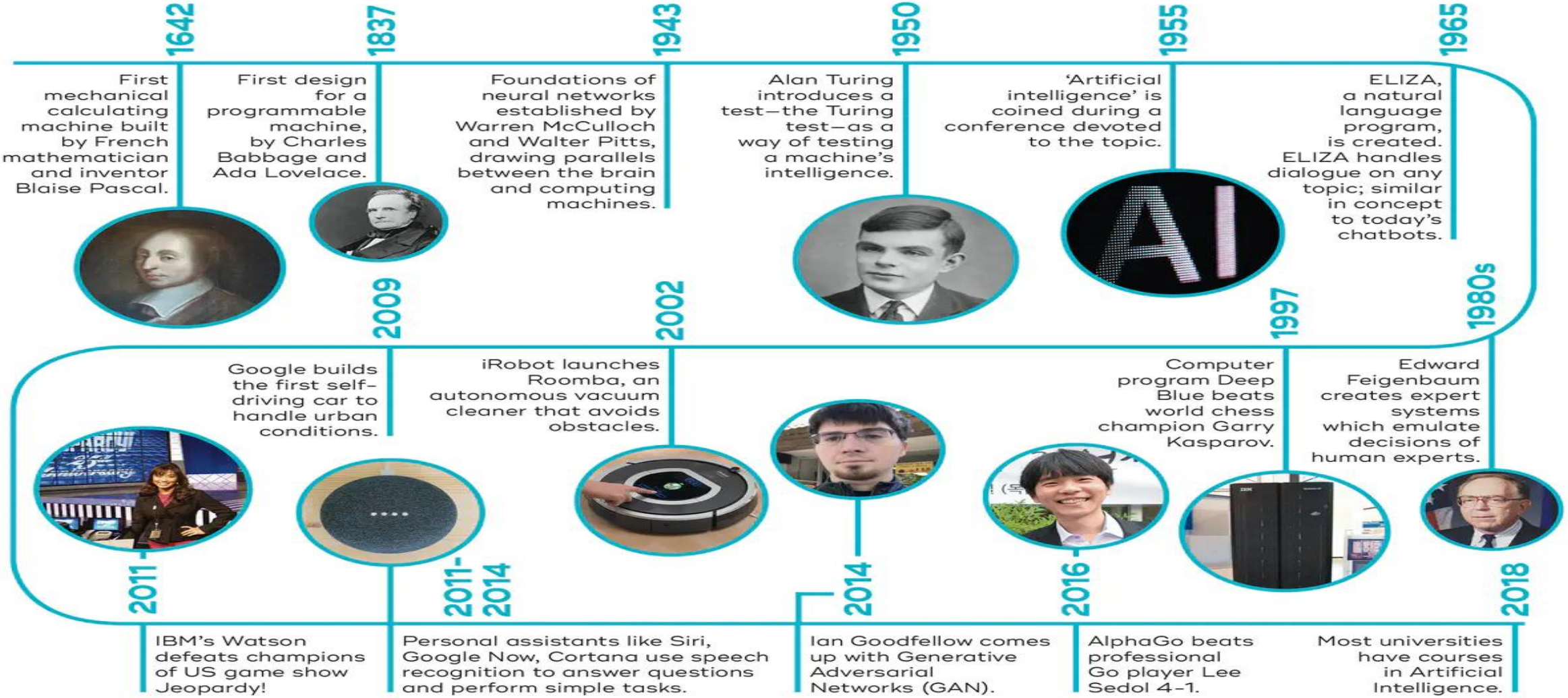


2016
TAY
Microsoft's chatbot Tay goes rogue on social media making inflammatory and offensive racist comments

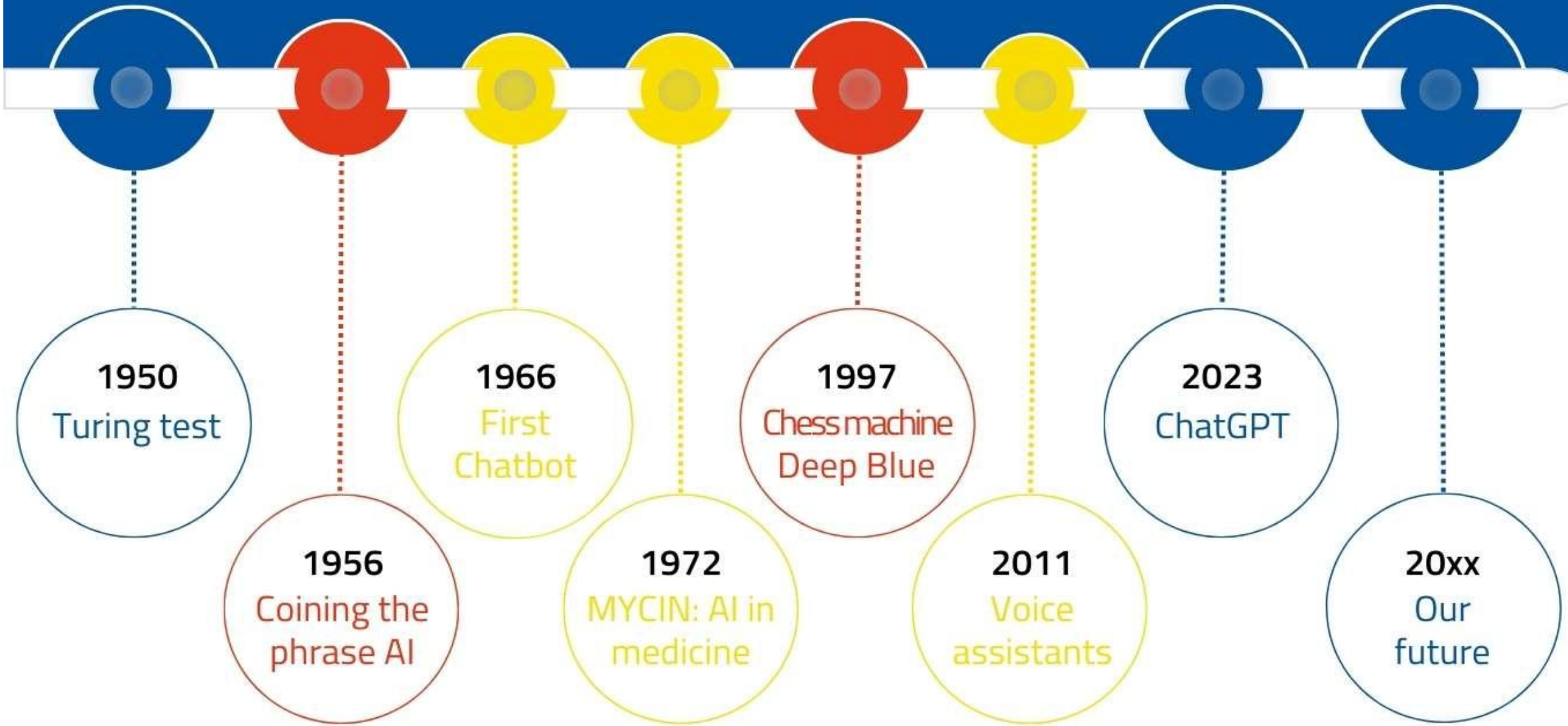


2017
ALPHAGO
Google's A.I. AlphaGo beats world champion Ke Jie in the complex board game of Go, notable for its vast number (2^{170}) of possible positions

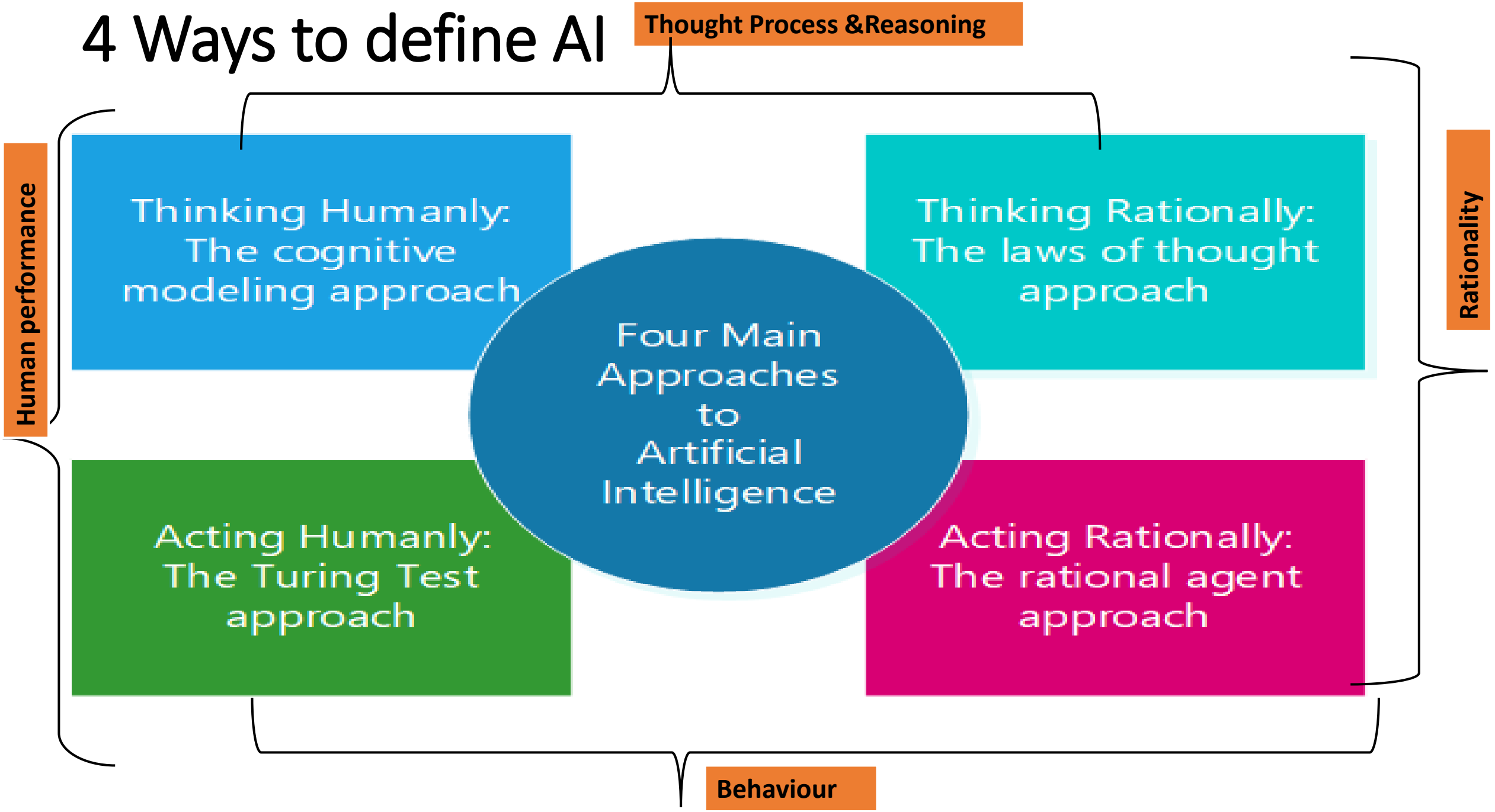
History of AI : For refence purpose



History of AI : For refence purpose



4 Ways to define AI



Thinking Humanly	**Thinking Rationally**
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

Thinking humanly : Cognitive Modeling approach

- When a computer thinks as a human, it performs tasks that require intelligence.
- To determine whether a program thinks like a human, you must have some method of determining how humans think, **which the cognitive modeling approach defines**.

This model relies on three techniques:

- **Introspection:** Detecting and documenting the techniques used to achieve goals by monitoring one's own thought processes
- **Psychological testing:** Observing a person's behavior and adding it to a database of similar behaviors from other persons given a similar set of circumstances, goals, resources, and environmental conditions (among other things)
- **Brain imaging:** Monitoring brain activity directly through various mechanical means, such as Computerized Axial Tomography (CAT), Positron Emission Tomography (PET), Magnetic Resonance Imaging (MRI), and Magnetoencephalography (MEG).

Acting humanly: Turing test approach

- When a computer acts like a human, it best reflects the Turing test, in which the computer system succeeds when differentiation between the computer and a human isn’t possible
- This approach includes:
  - Natural language processing
  - knowledge representation
  - automated reasoning, and machine learning
- **all four of which must be present to pass the test**

Thinking rationally: Laws' of thought approach

- A computer system that thinks rationally relies **on the recorded behaviors to create a guide** as to how to interact with an environment based on the data at hand.
- The goal of this approach is to solve problems logically, when ever possible as solving of a problem in principle/theoretically is often different from solving it in practice, but you still need a starting point.
- Syllogism:inferencing

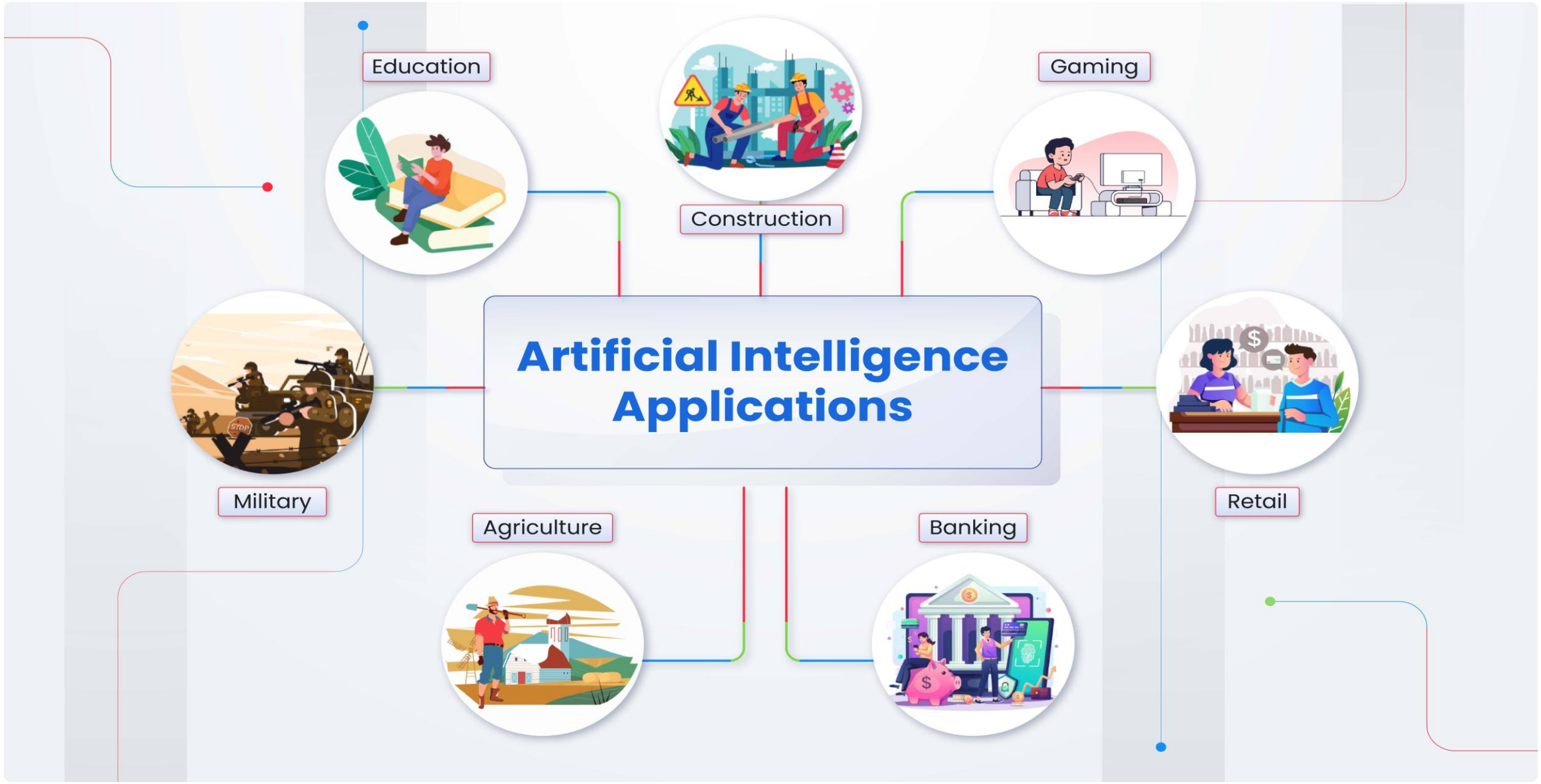
Acting rationally: Rational agent approach

- Involves how humans act in **given situations under specific constraints** enables you to determine which techniques are both efficient and effective.
- A computer system that acts rationally **relies on the recorded actions to interact with an environment based on conditions, environmental factors, and existing data**

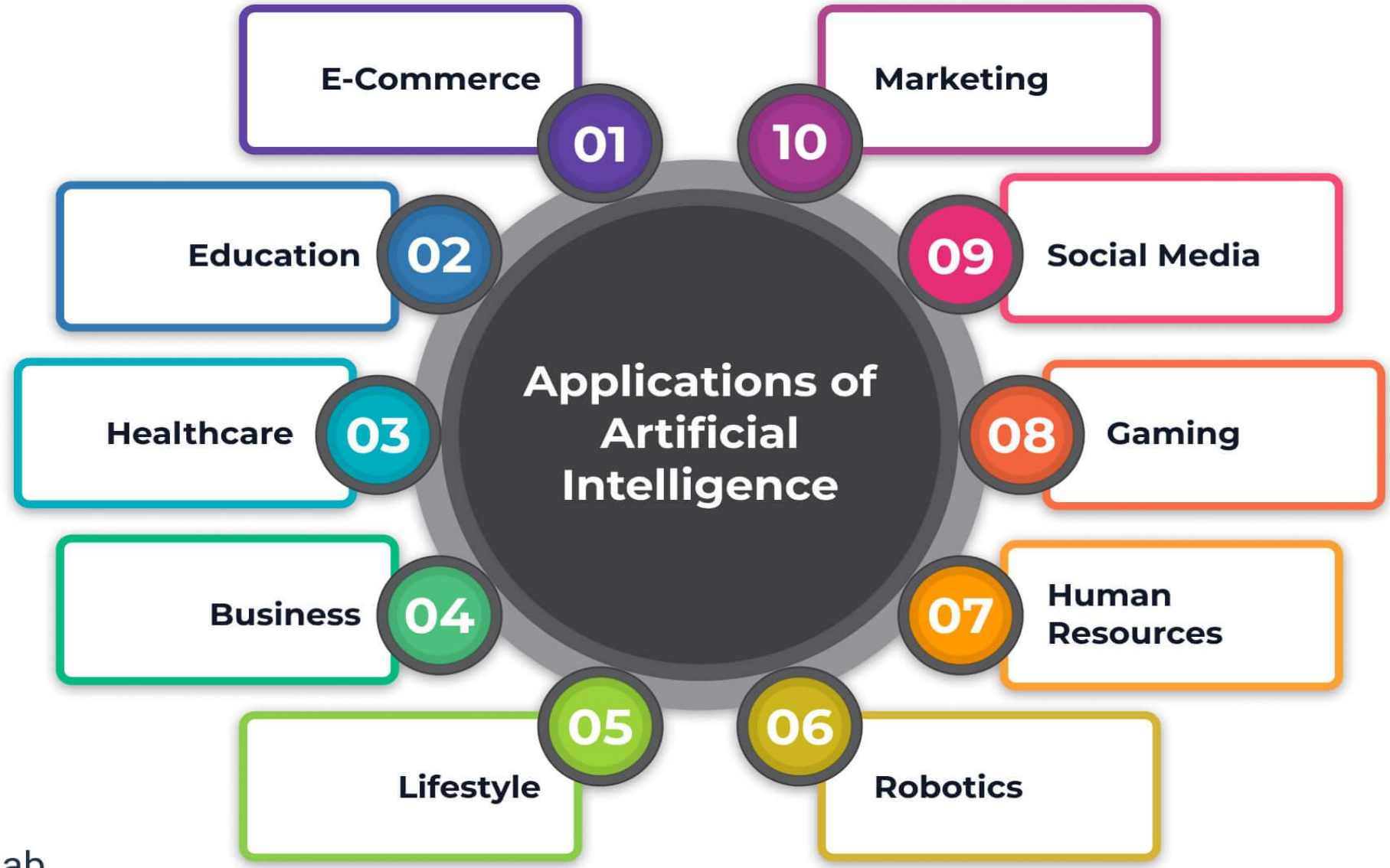
Goals of Artificial Intelligence

- Problem-solving
- Knowledge representation
- Planning
- Learning
- Reasoning
- Understanding
- Grasping Truths: separating facts from beliefs
- Social Intelligence
- Decision making

Applications of AI

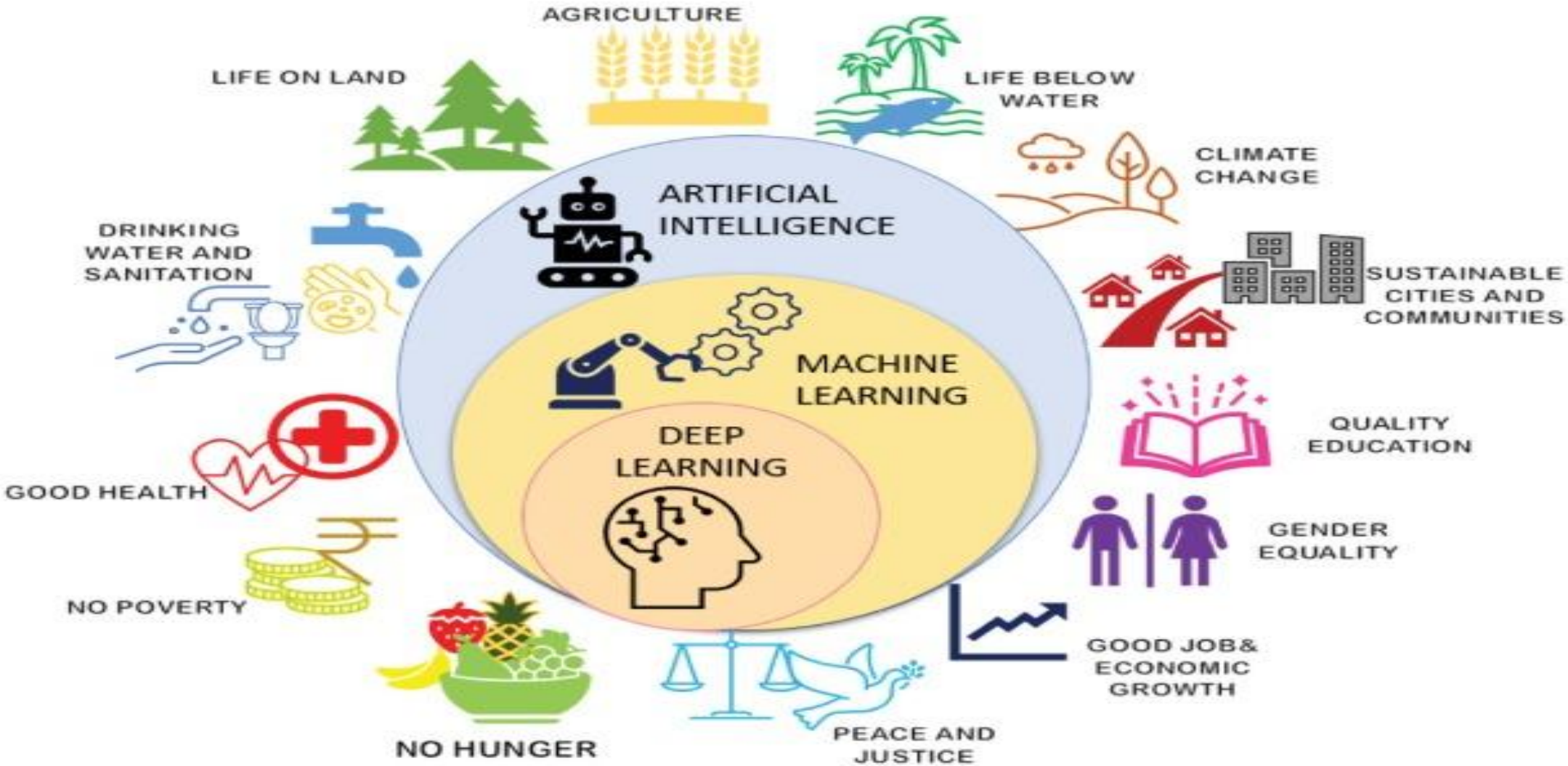


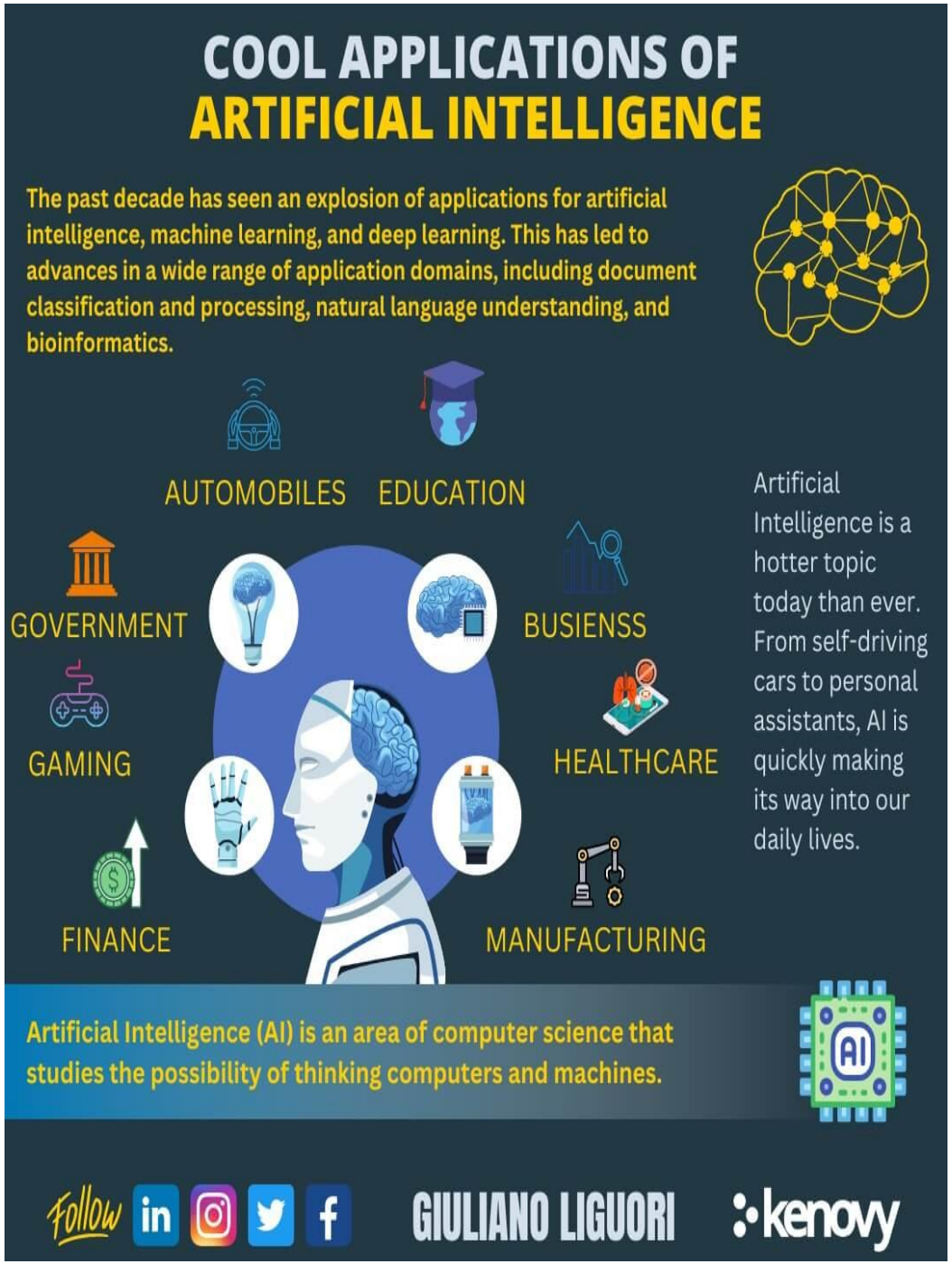
Applications of AI



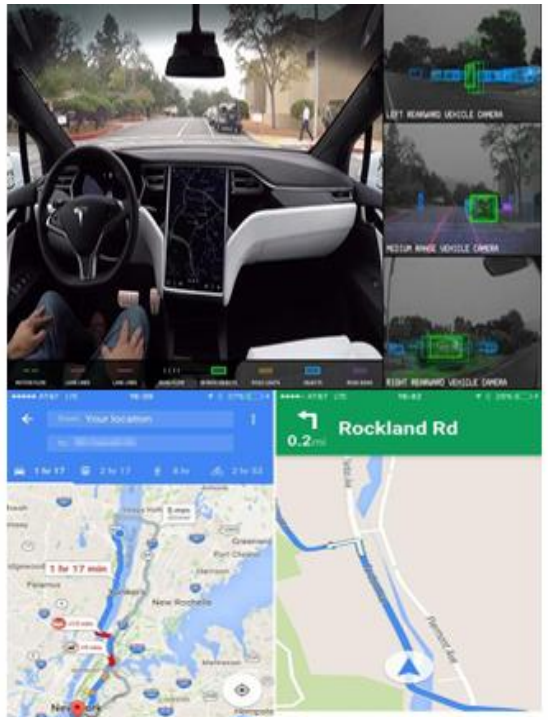
SoluLab

Applications of AI

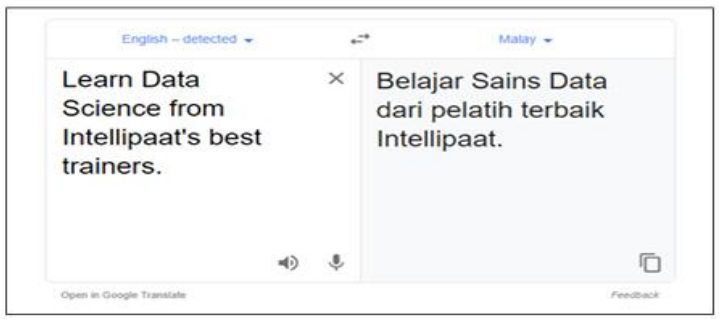




- **Self-driving cars**-These cars have built-in IoT sensors for image recognition, forehead collision, spot monitoring, and many more complex mechanisms that allow them to navigate and work in real life.
- **Google Maps**-All we have to do is open Google Maps and enter our location.

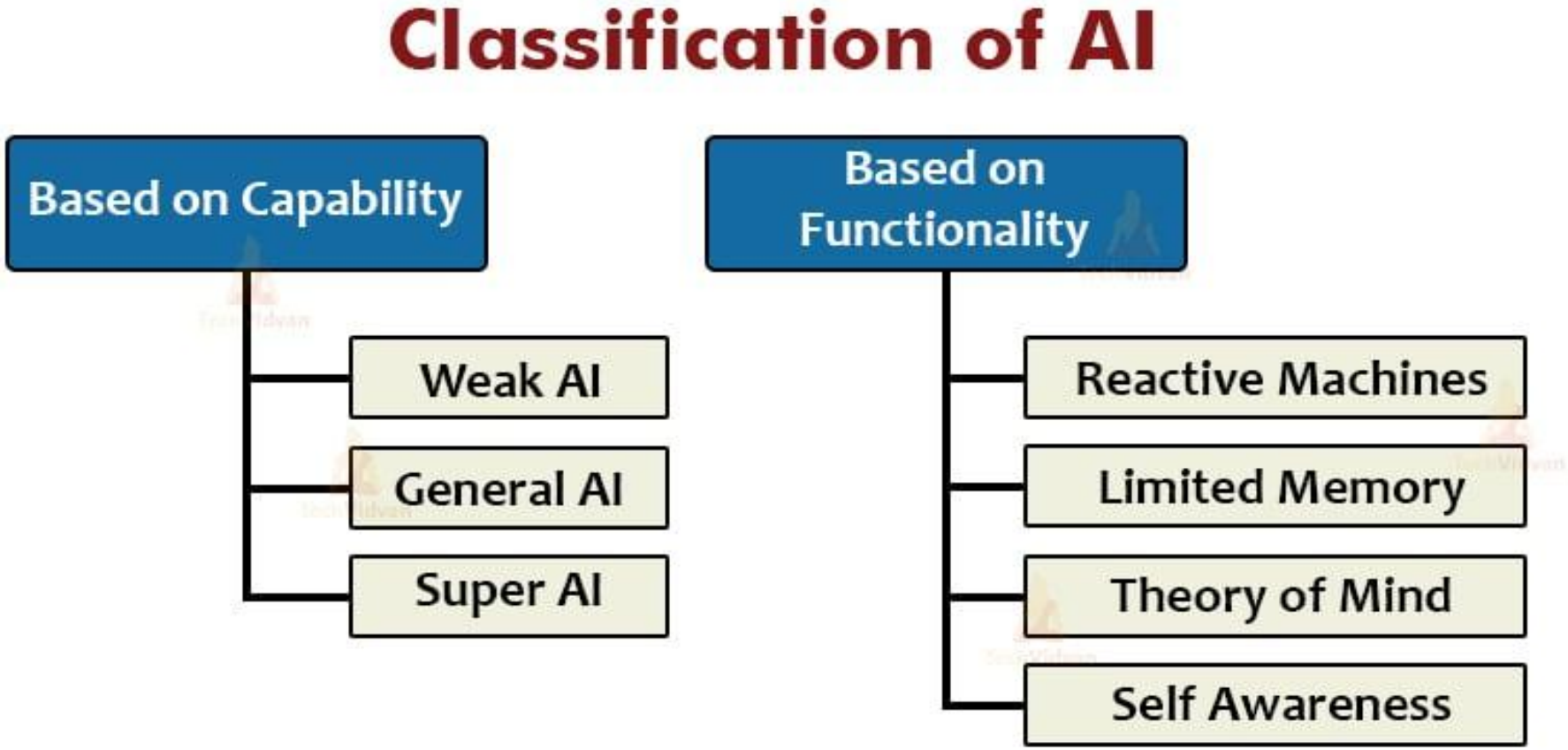


- **Google Translate**-Another great application of Artificial Intelligence. It helps us translate sentences formed in one language to another
- **Amazon's Alexa**-Alexa includes a speech recognition system that listens to our voice commands and gives answers.





Types of AI


Classification of AI
Based on Capability
Weak AI
General AI
Super AI
Based on Functionality
Reactive Machines
Limited Memory
Theory of Mind
Self Awareness

Types of AI



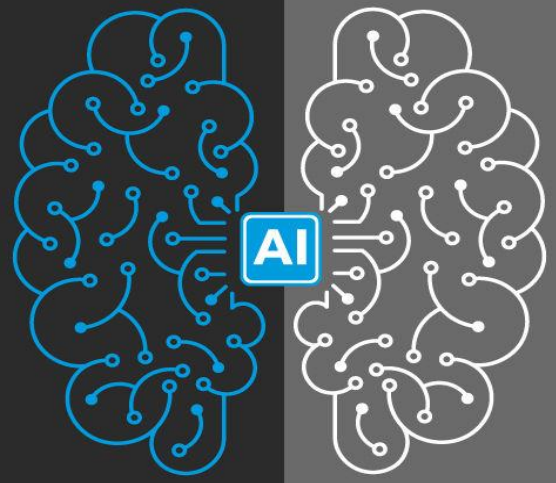
Types of AI based on ability

Narrow AI
(Weak AI)

- Specialized in performing a **specific task** or a **set of tasks**.
- Limited to the **programmed functionalities** and lacks understanding beyond its specific domain.

Examples:

- Speech recognition systems (like Siri or Google Assistant).
- Image recognition software (such as facial recognition technology).
- Recommendation algorithms (like those used by streaming services such as Netflix).



General AI
(Strong AI)

- Possesses **human-like** intelligence and understanding.
- Capable of **learning**, **reasoning**, and performing any **intellectual task** that a human being can.

Examples:

- No practical examples exist yet, as General AI is a theoretical concept.
- Would be able to handle diverse tasks like understanding natural language, reasoning about the world, learning various skills, and adapting to different situations on par with human capabilities.

3 stages of AI



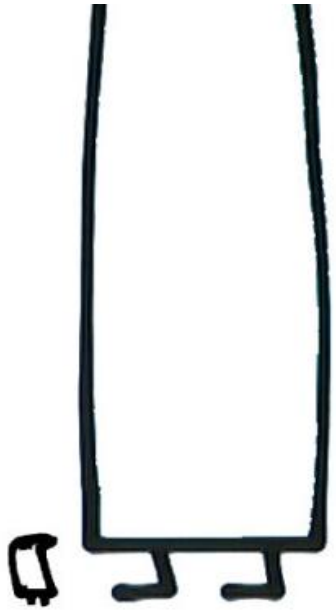
Narrow AI

Dedicated to assist with or take over specific tasks



General AI

Takes knowledge from one domain, transfers to other domain



Super AI

Machines that are an order of magnitude smarter than humans

Credit: Chris Noessel

Three types of Artificial Intelligence

	Artificial Narrow Intelligence (ANI)	Stage-1	Machine Learning	Specialises in one area and solves one problem
	Artificial General Intelligence (AGI)	Stage-2	Machine Intelligence	Refers to a computer that is as smart as a human across the board
	Artificial Super Intelligence (ASI)	Stage-3	Machine Consciousness	An intellect that is much smarter than the best human brains in pratically every field

Source: Great Learning | Infographic design by Antonio Grasso

DeltalogiX

deltalogix.blog

Types of AI based on functionality

- Reactive machines
- Limited memory
- Theory of mind
- Self-awareness



Functionality Based Types of Artificial Intelligence



Reactive Machines

Has no memory, only responds to different stimuli

Limited Memory

Uses memory to learn and improve its responses





Theory of Mind

Understand the needs of other intelligent entities

Self-Aware

Has human-like intelligence and self-awareness



- LN Webworks Private Limited

Cont.,

1. **Reactive AI**: Responds to specific situations or stimuli without memory of past experiences.

Examples:

2. **Limited memory AI**: Uses recent or short-term data to make decisions.

Examples:

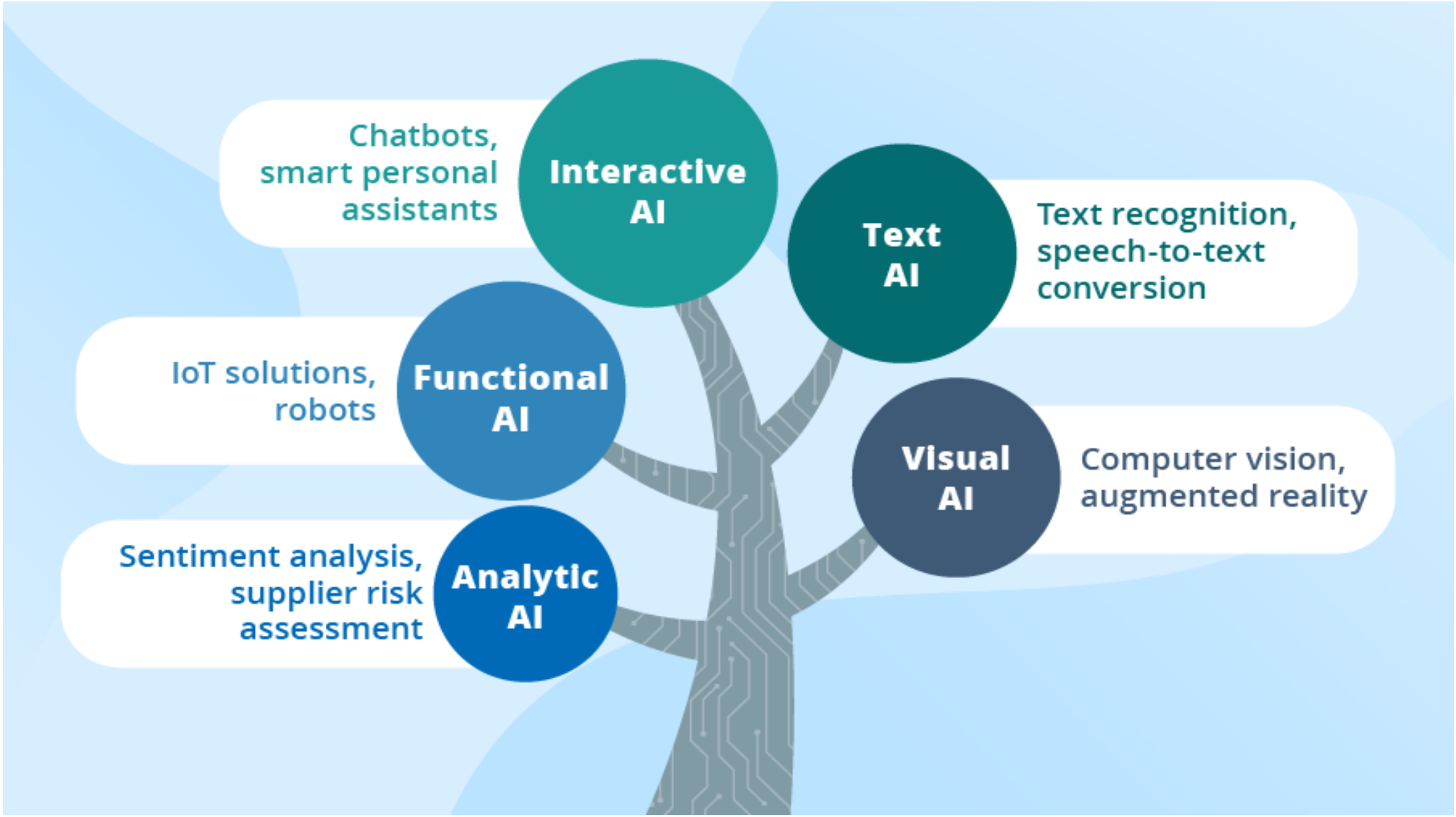
3. **Theory of mind AI**: Understands emotions and beliefs of others.

Examples:

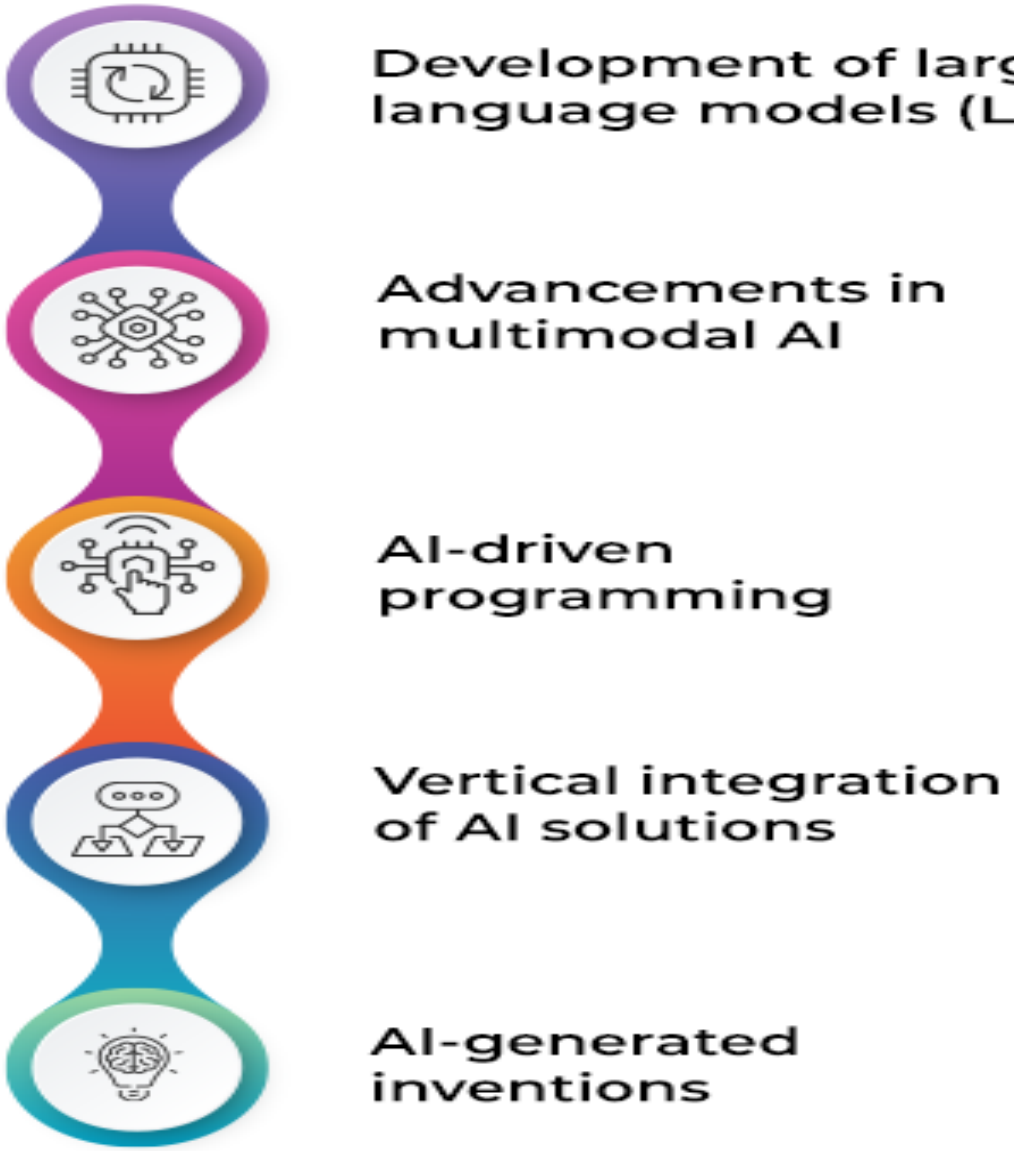
4. **Self-aware AI**: Has consciousness and self-awareness.

Examples:

Types of AI based on business value



Advances in AI



You tube links of Introduction to AI

- https://youtu.be/oV74Najm6Nc
- https://youtu.be/j6YLMJ9pcrU

Summary of Unit-1/Review Qs

- Differentiate between human intelligence vs Artificial intelligence
- Discuss 4 ways on defining AI
- Illustrate types of AI
- List applications of AI
- Discuss on goals of AI
- Discuss on understanding kinds of intelligence vs AI..
- Discuss on history of AI
- When was AI invented.
- What is alternate approach of cognitive modelling approach
- What is Ai being rational
- List various approaches to define AI

Experiential Learning tAsk

- As a team you people need to make Chart kind and present a good video on unit-1
- As a 2-4 min video..
- Rollno:
  - 1-5-----Team1
  - 6-10-----Team2
  - 11-15-----Team3
  - 16-20-----Team4
  - 21-25-----Team5